

Gamified Personal Tracker System - Detailed Design

1. Overview

A system to track daily activities and convert them into a gamified experience using XP, levels, streaks, and achievements.

2. System Architecture

- 1 Frontend: React / Next.js
- 2 Backend: Spring Boot (Java 17)
- 3 Database: PostgreSQL
- 4 Optional: Redis (for caching streaks, sessions)

3. Core Features

- 1 Activity tracking (start/stop, manual entry)
- 2 Daily dashboard
- 3 XP and leveling system
- 4 Streak tracking
- 5 Achievements

4. Data Model

Activity(id, name, category)

LogEntry(id, activity_id, start_time, end_time, duration)

Skill(id, name, xp, level)

XPLog(id, activity_id, xp_earned, timestamp)

Achievement(id, name, description, condition)

5. Gamification Logic

XP Formula: $XP = \text{time_spent_minutes} * \text{multiplier}$

Level Formula: $XP_required = 100 * (\text{level} ^ 1.5)$

Multipliers example:

- 1 Study: 1.5x
- 2 Work: 1.2x
- 3 Chores: 1.0x
- 4 Gaming: 0.8x

6. Streak System

- 1 Track daily consistency
- 2 Reward bonus XP for streak continuation
- 3 Break streak resets bonus

7. Achievements

- 1 First activity logged
- 2 10 hours study milestone
- 3 7-day streak
- 4 High productivity day

8. API Design (Sample)

- 1 POST /activities
- 2 POST /log/start
- 3 POST /log/stop
- 4 GET /dashboard
- 5 GET /skills

9. Future Enhancements

- 1 AI insights (best productivity time)
- 2 Smart XP boosts
- 3 Quests and challenges
- 4 Leaderboards